A4.1 Evolution and speciation (HL)

NAME	DATE	TOTAL
		_____ / 20

About 20 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

5 MARKS

1 DEFINE allopatric speciation. [A4.1.8] [1]

2 DEFINE adaptive radiation. [A4.1.9] [1]

3 STATE one behaviour that often prevents hybridization in animals. [A4.1.10] [1]

4 DEFINE polyploidy. [A4.1.11] [1]

5 STATE the term for a polyploid that contains chromosome sets from two species. [A4.1.11] [1]

Section B: Understanding

7 MARKS

6 COMPARE allopatric and sympatric speciation. [A4.1.8] [3]

7 EXPLAIN how adaptive radiation increases biodiversity. [A4.1.9] [2]

8 EXPLAIN how sterile hybrids stop alleles passing from one species to another. [A4.1.10] [2]

Section C: Working with evidence

8 MARKS

All data in this section are invented for practice. The patterns are realistic.

9 [A4.1.11]

Three related plant species grow in the same marsh. Species C has features of both A and B.

SPECIES	A	B	C
CHROMOSOME NUMBER IN BODY CELLS	22	22	44
NUMBER OF CHROMOSOMES IN ONE BASIC SET	11	11	11

a DEDUCE how many sets of chromosomes species C has. [1]

b EXPLAIN how species C could have formed from A and B. [3]

c EXPLAIN why species C is reproductively isolated from species A. [2]

d STATE why this kind of speciation is described as abrupt. [1]

e SUGGEST one advantage that species C might have over its parents. [1]